

CHRONIC LUNG DISEASE PREDICTIVE ANALYTICS

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CHRONIC LUNG DISEASE PREDICTIVE ANALYTICS

A Project Report submitted in partial fulfillment of the

requirements for the award of the degree of

**Bachelor of Technology
in
Computer Science & Engineering**

by

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This is to certify that the project report entitled “**Chronic Lung Disease Predictive Analytics**” submitted by **Pakala Sriman Reddy, Poonam Suthar, P. Shreemayi** to the Institute of Aeronautical Engineering, Hyderabad in partial fulfillment of the requirements for the award of the Degree Bachelor of Technology in Computer Science and Engineering is a bonafide record of work carried out by him/her under my/our guidance and supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute for the award of any Degree.

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ABSTRACT

Keywords: COPD, Machine Learning, Deep Belief Network, Predictive Analytics, Django, Python, Chatbot, EHR, Respiratory Sound, Healthcare AI.

Chronic Obstructive Pulmonary Disease (COPD) is a progressive and preventable condition that significantly impacts global health, posing serious socio-economic challenges due to its high rates of morbidity and mortality. This project, developed in collaboration with **Teladoc Health**, focuses on building a predictive analytics platform that utilizes machine learning to assess the risk of COPD and assist in early diagnosis and intervention. Using supervised Deep Belief Network (DBN) models, the system analyses respiratory audio samples uploaded by users to classify them as either "COPD" or "Healthy." The platform also includes a chatbot to provide users with disease-related information and educational support.

The system architecture integrates a Python backend using Django, a frontend developed with HTML, CSS, and JavaScript, and an MSSQL database. The predictive model was trained on a labelled dataset sourced from Kaggle. The application supports both admin and user roles, where admins manage doctors and feedback, and users can register, upload sound files for prediction, and receive guidance. The project also envisions the integration of Electronic Health Records (EHRs), wearable sensors, and real-time environmental data to personalize treatments and monitor patient progress effectively. This system not only assists in prediction but also aims to transform COPD management by offering real-time data insights, contributing to improved public health outcomes.

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Chronic Obstructive Pulmonary Disease (COPD) is a long-term, progressive respiratory disease that primarily causes airflow limitation and breathing difficulties. It is characterized by chronic inflammation in the lungs, which can result from prolonged exposure to harmful particles such as tobacco smoke, environmental pollutants, and industrial dust. According to the World Health Organization, COPD is among the leading causes of death worldwide and is expected to become the third most common cause of mortality by 2030. The rising prevalence of this disease underscores the urgent need for innovative and proactive healthcare solutions focused on early diagnosis and intervention.

In response to this pressing healthcare challenge, Teladoc Health has supported the development of an intelligent system—Chronic Lung Disease Predictive Analytics—to assist in the early detection of COPD using machine learning (ML) technologies. This project leverages a Supervised Deep Belief Network (DBN) classification algorithm to analyse respiratory sound data and determine whether a user falls into the "COPD" or "Healthy" category. The system is designed to predict acute exacerbations before they occur, enabling timely medical attention and better management of the condition. It also integrates a chatbot to help users understand more about the diagnosed condition, increasing awareness and patient engagement.

The platform provides separate access for users and administrators. Users can register, log in, upload their respiratory sound recordings, receive predictive results, and consult area-wise filtered doctor listings. Admins can manage doctors, monitor feedback, and oversee user activity. The system is built using Python with the Django framework for the backend, while the frontend employs HTML, CSS, and JavaScript. Data is stored and managed using an MSSQL database, and the model is trained on a respiratory sound dataset sourced from Kaggle.

This project represents a significant step toward applying artificial intelligence to real-world healthcare problems. By enabling early risk prediction and combining insights from Electronic Health Records (EHR), wearable devices, and environmental data, the system offers a holistic view of patient health. Furthermore, it provides a scalable, cost-effective solution that can be deployed across clinics, telehealth platforms, and personal health applications. The primary objective is not only to predict and prevent COPD-related complications but also to support personalized treatment strategies and improve long-term patient outcomes.

1.2 EXISTING SYSTEM

In the current healthcare landscape, the diagnosis and management of Chronic Obstructive Pulmonary Disease (COPD) primarily rely on traditional clinical assessments and spirometry tests conducted in hospitals or clinics. These methods, although standard, often detect the disease

only in its advanced stages when lung damage has already occurred. Additionally, existing systems lack real-time monitoring capabilities and do not leverage modern technologies like artificial intelligence or machine learning to assist in early prediction or automated diagnosis. Moreover, there is a noticeable absence of patient-centric tools that allow individuals to assess their risk remotely or receive timely alerts about potential exacerbations. While some platforms provide general health tracking and electronic health record (EHR) management, they are not specifically designed for respiratory disease prediction or do not integrate predictive analytics based on physiological inputs like respiratory sound data. These systems also typically do not offer chatbot support to help educate patients or guide them post-diagnosis, limiting their utility in day-to-day health management.

Furthermore, most existing solutions do not offer a centralized platform where users can easily find specialist doctors, provide feedback, or interact with a responsive interface tailored to respiratory health. This lack of technological integration and user support mechanisms creates a gap in providing effective, scalable, and accessible COPD management solutions.

1.3 DEMERITS OF EXISTING SYSTEM

The current systems for diagnosing COPD primarily rely on manual clinical assessments and spirometry, often detecting the disease only at advanced stages. They lack machine learning-based predictive models, limiting early intervention opportunities. Real-time monitoring is rarely available, and there's minimal integration with wearable devices or electronic health records. Additionally, users cannot interact with systems for guidance, as most lack chatbot support. Finding specialized doctors is often inconvenient, and there's typically no feedback mechanism to improve user experience. Overall, these systems are not user-centric and fail to utilize modern technologies for efficient disease management.

1.4 PROPOSED SYSTEM

The system leverages recent advancements in deep learning, where models like DeepSpiro and AI-PFT-Clin have demonstrated high accuracy (AUC up to 0.83–0.84) in predicting COPD and its exacerbations by analyzing spirometric curves, demographic data, and clinical features. These frameworks highlight the importance of integrating heterogeneous data streams—such as lung sounds, spirometry, and patient history—for robust prediction and early intervention.

Moreover, multimodal deep learning frameworks, such as COPD-MMDDxNet, further enhance diagnostic accuracy and accessibility, especially in resource-constrained settings, by combining multiple data sources and optimizing feature extraction. The proposed platform's design aligns with these trends, supporting both early detection and longitudinal risk assessment for COPD, thereby enabling timely and personalized healthcare interventions.

Additionally, recent studies emphasize the clinical interpretability and explainability of deep learning models, which are crucial for gaining trust among healthcare providers and patients. By incorporating explainable AI (XAI) techniques, the system can provide transparent predictions and actionable insights, improving clinical decision-making and patient engagement.

1.5 MERITS OF PROPOSED SYSTEM OVER EXSISTING SYSTEM

- **Early Prediction with AI:** By using Deep Belief Network (DBN) algorithms, the system predicts COPD in its early stages, enabling timely intervention and improved patient outcomes.
- **Automation and Intelligence:** The system intelligently analyses audio input to detect symptoms, reducing dependence on manual diagnosis and enhancing accuracy.
- **User-Friendly Interface:** A clean and responsive front-end allows users to easily interact with the system, upload sound files, and navigate doctor listings without technical barriers.
- **Integrated Chatbot Support:** The chatbot feature helps users understand their diagnosis, receive personalized advice, and learn more about COPD, improving awareness and engagement.
- **Doctor Discovery Based on Location:** The platform allows users to search for doctors based on geographical filters, streamlining access to medical support.
- **Admin Functionality:** Administrators can manage users, doctors, and feedback, ensuring smooth operation and timely content updates.
- **Feedback Mechanism:** Unlike existing systems, users can submit feedback, enabling continuous improvement of the platform and its services.
- **Real-Time Monitoring Potential:** With scope for integration with wearable devices and EHRs, the system supports continuous health monitoring and data-driven decision-making.
- **Scalable and Secure:** Developed with Django and backed by a secure database, the system ensures user data is handled safely and can be scaled for wider deployment.

1.6 REQUIREMENTS

1.6.1 SOFTWARE REQUIREMENTS

- **Operating System:** Windows, Linux, or macOS

- Programming Languages: Python (for backend development), JavaScript (for frontend), HTML, CSS
- Frameworks: Django (for backend), Flask (for certain modules), TensorFlow/PyTorch (for machine learning)
- Database: MSSQL or PostgreSQL for secure data storage
- Machine Learning Libraries: Scikit-learn, TensorFlow, PyTorch (for DBN-based prediction models)
- Chatbot Integration: Libraries such as Rasa or Dialogflow for chatbot implementation
- Web Technologies: HTML5, CSS3, JavaScript, and Bootstrap for frontend development
- Server: Apache or Nginx for hosting
- Cloud Integration: AWS or Azure for cloud hosting and deployment
- Version Control: Git for source code management
- Authentication: OAuth or JWT for secure login and session management

1.6.2 HARDWARE REQUIREMENTS

- Server:

The server, either cloud-based or local, should have at least 16 GB RAM and 500 GB storage to host the application and store necessary data like user profiles, sound files, and model results. Cloud services like AWS, Azure, or Google Cloud can be used for scalability and reliability.

- Client Machines:

Users should access the system from devices with 4 GB RAM or more, using modern browsers such as Chrome, Firefox, or Safari. A stable internet connection is required for uploading sound files and receiving results.

- Wearable Devices :

Optional integration with wearable health devices like Fitbit or Apple Watch can provide real-time health data (e.g., heart rate, oxygen levels) to enhance predictive accuracy.

- Audio Input Device:

Microphones on devices like smartphones or computers are needed to record sound files for COPD detection, with clear audio capture essential for accurate analysis.

CHAPTER 2

LITERATURE SURVEY

2.1 Detection of Explosive Cough Events in Audio Recordings

<https://ieeexplore.ieee.org/abstract/document/8037429>

- Purpose: To develop an internal sound analysis method for detecting sudden, intense cough events ("explosive" coughs).
- Context: With respiratory conditions like COPD and asthma, cough characteristics offer critical diagnostic clues.
- Method: Signal processing techniques analyze features like intensity, duration, and spectral shape of coughs in real-world audio.
- Significance: Supports integration into automated health monitoring apps and early detection systems, especially in remote healthcare scenarios.

2.2 Validation of Automatic Wheeze Detection in Obstructive Airway Diseases

<https://www.tandfonline.com/doi/abs/10.1080/02770900802386008>

- Goal: Validate an algorithm that automatically detects wheezing sounds in patients and compares performance with human experts.
- Background: Wheezing is a critical symptom in asthma and COPD; accurate detection is essential for diagnosis and monitoring.
- Approach: Applied the algorithm on recordings from both healthy and obstructed airways.
- Result: Strong correlation with clinical assessments, demonstrating high diagnostic reliability.
- Potential: Useful in telemedicine and continuous health monitoring environments.

2.3 Creation of a Respiratory Sound Database for Classification

https://link.springer.com/chapter/10.1007/978-981-10-7419-6_6

- Purpose: To provide a standardized dataset of annotated respiratory sounds from various patients.
- Details: Includes normal breath sounds, wheezes, crackles, and rhonchi, collected under diverse clinical conditions.
- Importance: Enables benchmarking and training of machine learning models for lung sound classification.

- Outcome: This dataset is now one of the key resources used in respiratory audio AI research and competitions (e.g., PhysioNet).

2.4 Detection of Lung Status Using Morphological Complexity of Respiratory Sounds

<https://onlinelibrary.wiley.com/doi/full/10.1155/2014/182938>

- Objective: Assess lung health by analyzing the complexity of lung sound signals using fractal and nonlinear analysis.
- Concept: Diseased lungs tend to produce more chaotic or complex sound patterns compared to healthy lungs.
- Methodology: Extracted features like Higuchi's fractal dimension and permutation entropy to distinguish signal irregularities.
- Findings: The proposed features proved effective in categorizing lung conditions with higher accuracy.
- Implication: Promotes use of advanced mathematical techniques in audio diagnostics beyond conventional frequency-based analysis.

2.5 Application of Semi-Supervised Deep Learning to Lung Sound Analysis

<https://ieeexplore.ieee.org/document/7590823>

- Objective: To apply semi-supervised deep learning techniques for the automatic classification of lung sounds.
- Challenge: Labeled medical datasets are often limited; semi-supervised methods help leverage both labeled and unlabeled data.
- Approach: Combines Convolutional Neural Networks (CNNs) with manual feature extraction to improve accuracy with minimal annotation effort.
- Impact: Demonstrated that even with small labeled datasets, the deep learning model can achieve performance close to supervised models.
- Use Case: Suitable for deployment in low-resource settings or mobile health applications.

2.6 An Expert Diagnostic System for Asthma and COPD Detection

<https://www.nature.com/articles/s41598-018-30116-2>

- Purpose: To design an expert system capable of distinguishing between asthma and COPD based on clinical features.
- Techniques Used: Uses artificial intelligence (AI) techniques such as decision trees, rule-based classifiers, and respiratory sound patterns.

- Result: The system achieved high accuracy in clinical settings by analyzing patient symptoms, medical history, and auscultation data.
- Significance: Automates the diagnostic process, potentially reducing dependency on specialist evaluation.
- Future Scope: Integration into wearable devices or mobile platforms for continuous remote monitoring.

2.7 Automatic Adventitious Respiratory Sound Analysis

<https://pmc.ncbi.nlm.nih.gov/articles/PMC5446130/pmc.ncbi.nlm.nih>

- Purpose: To automate detection and classification of abnormal respiratory sounds like wheezes and crackles for better pulmonary diagnosis. Human auscultation varies, so AI offers standardized, objective analysis.
- Context: Respiratory sounds are recorded in noisy clinical settings, challenging accurate diagnosis. Deep learning with spectrogram features improves performance, aiding telemedicine and monitoring.
- Challenges: Differentiating overlapping sounds and handling noise lowers accuracy. Limited labeled data and variability in recordings further complicate classification.
- Solutions: CNNs with pretrained feature extraction on Mel spectrograms enhance detection accuracy. Transfer learning boosts performance beyond traditional methods and human experts.
- Feedback: Models achieved about 86.5% accuracy and AUC 0.93 for abnormal sounds. AI reduces variability due to human skill differences, standardizing assessments.
- Objective: Develop a reliable AI tool for early respiratory disease detection from auscultation sounds, supporting clinicians and enabling remote care applications

2.8 Respiratory Sound Classification for Crackles, Wheezes, Rhonchi

<https://www.nature.com/articles/s41598-021-96724-7nature>

- Objective: Deep learning classification on real clinical recordings with noise; abnormal detection accuracy $\approx 86.5\%$ and AUC ≈ 0.93 .
- Significance: Shows feasibility of triage-level screening using spectrogram-based CNNs in routine clinical environments.

2.9 ML for Audio-Based Respiratory Screening (Survey)

<https://pmc.ncbi.nlm.nih.gov/articles/PMC9791302/pmc.ncbi.nlm.nih>

- Goal: Survey of ML methods for remote auscultation (datasets like ICBHI, Coswara; CNN/RNN/Transformer models; noise and device variability challenges).

- Importance: Guides dataset selection, preprocessing, and robustness strategies for telemedicine deployments.

2.10 Recognition of Pulmonary Diseases from Lung Sounds (Deep Learning)

<https://pmc.ncbi.nlm.nih.gov/articles/PMC8019351/pmc.ncbi.nlm.nih>

- Purpose: Evaluate deep learning on ICBHI and locally recorded datasets for disease recognition from lung sounds.
- Note: Describes using ICBHI'17 cycles and end-to-end models, informing data curation and model design.

2.11 Automatic Lung Health Screening Using Respiratory Sounds

<https://pmc.ncbi.nlm.nih.gov/articles/PMC7797201/pmc.ncbi.nlm.nih>

- Objective: Build a screening tool using cepstral features and MLP on ICBHI; discusses multi-condition data and device variability.
- Relevance: Useful baseline for traditional feature-engineering vs deep models and for dataset/statistics on ICBHI cycles.

2.12 Deep Learning Survey: Lung Sound Classification

<https://www.internationaljournalssrg.org/IJECE/2025/Volume12-Issue4/IJECE-V12I4P120.pdfinternationaljournalssrg>

- Objective: Survey highlighting CNN/CRNN/attention models outperforming classical ML when sufficient data/augmentation are available.
- Use: Supports choosing deep models or semi-supervised variants for limited-label scenarios

2.13 Audio-Based Digital Biomarkers for Respiratory Diseases

<https://pmc.ncbi.nlm.nih.gov/articles/PMC12076160/pmc.ncbi.nlm.nih>

- Purpose: Systematic review on diagnostic accuracy of audio biomarkers for asthma, COPD, pneumonia; methods, datasets, and bias assessment.
- Implication: Frames clinical validation requirements and reporting for future translation.

2.14 Pediatric Lung Sound Analysis with AI

<https://www.jmir.org/2025/1/e66491jmir>

- Focus: Pooled performance of ML models for pediatric lung sound tasks, outlining data/labeling challenges and model trends.
- Relevance: Useful if pediatric cohorts or generalization across ages are in scope.

2.15 Cough Sounds in Screening and Diagnostics

https://stacks.cdc.gov/view/cdc/150417/cdc_150417_DS1.pdfstacks.cdc

- Scope: Reviews cough-based diagnostics across diseases, datasets, and model pipelines; emphasizes smartphone-based feasibility.
- Use Case: Complements explosive cough detection with broader clinical applications and data collection guidance.

CHAPTER 3

METHODOLOGY AND IMPLEMENTATION

3.1 METHODOLOGY

The methodology for developing the Chronic Lung Disease Predictive Analytics system involves the following key steps:

3.1.1 Data Collection:

The dataset for COPD prediction is collected from Kaggle, which includes various features like patient demographics, medical history, and environmental factors. Audio recordings of the patient's cough are also used as part of the input to the prediction model.

3.1.2 Preprocessing:

Data cleaning is performed to handle any missing values, inconsistencies, or noise in the dataset. Normalization or standardization may be applied to ensure the model trains efficiently. Audio data is preprocessed by converting sound files into feature vectors, which are then analyzed by the machine learning model for COPD prediction.

3.1.3 Feature Selection:

Relevant features for predicting COPD are selected based on their importance and correlation with the disease. Features like smoking history, air quality, and age play a significant role in predicting the severity of the disease.

3.1.4 Model Training:

A Supervised Deep Belief Network (DBN) classification algorithm is employed to train the model on the dataset. DBNs are ideal for this task due to their ability to model complex relationships and deal with the vanishing gradient problem in deep networks. The model is trained to classify patients as either "COPD" or "Healthy" based on the available features.

3.1.5 Model Evaluation:

The performance of the model is evaluated using metrics such as accuracy, precision, recall, and F1-score. Cross-validation is used to ensure the model generalizes well to unseen data.

3.1.6 Deployment:

After model evaluation, the trained model is integrated into the Django-based web application, where users can upload sound files (e.g., cough recordings) to receive a prediction regarding

their risk of COPD. The system also includes a chatbot that provides further information about COPD and answers user queries.

3.1.7 Continuous Improvement:

User feedback is gathered through the system's interface, and it helps improve the accuracy and usability of the model. The system is also designed to integrate additional data sources, such as wearable devices or real-time environmental data, for continuous health monitoring.

3.2 SYSTEM ARCHITECTURE

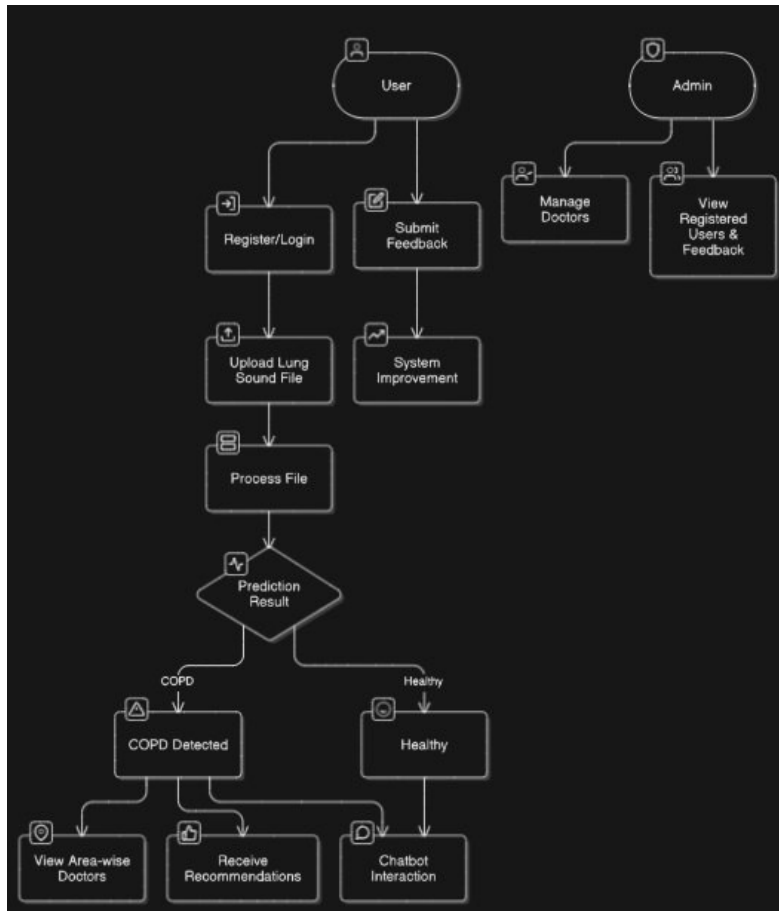


Figure 1: System architecture

a. User Interface (Frontend):

The frontend is developed with HTML, CSS, and JavaScript, ensuring responsiveness across devices to serve diverse users. It includes secure user registration and login, enabling personalized profiles and data privacy. The file upload feature supports multiple audio formats for cough recordings, with progress feedback for smooth user experience. The doctor

search page filters listings by area, improving accessibility for clinical consultations. A chatbot provides real-time interactive guidance about COPD symptoms, preventive advice, and next steps post-prediction, enhancing user engagement and health literacy.

b. Backend:

Built on Python's Django framework, the backend ensures robust security for user authentication and session management. It handles audio preprocessing—noise reduction, segmentation—and extracts relevant acoustic features for input to the DBN model. The prediction pipeline efficiently manages asynchronous inference, returning results with low latency. It also logs user feedback and doctor interactions into the system, supporting continuous improvement. APIs facilitate communication between frontend components, database, and ML services, enabling scalable and maintainable system architecture.

c. Database:

Using Microsoft SQL Server, the database securely stores user credentials, profile details, and activity logs with encryption to protect sensitive data. It maintains historical records of predictions and user feedback, supporting longitudinal patient monitoring and research. The doctor directory includes geolocation data for efficient filtering by user area and can be updated securely by administrators. Data indexing and optimization ensure fast query responses, facilitating smooth user experience for both patients and healthcare providers.

3.3 ALGORITHM

A Deep Belief Network (DBN) is an advanced machine learning model based on a stack of Restricted Boltzmann Machines (RBMs), which learns hierarchical representations of input data. This architecture is well-suited for predicting Chronic Obstructive Pulmonary Disease (COPD) based on medical data and audio input (e.g., cough recordings). Below is the detailed explanation of the DBN algorithm for COPD prediction.

3.3.1 Architecture of DBN

- **Input Layer:** Takes medical features (e.g., age, smoking history) and audio features from cough recordings (e.g., pitch, intensity).
- **Hidden Layers (RBMs):** Each RBM layer learns complex, abstract features from the input data in an unsupervised manner.
- **Output Layer:** A softmax layer that classifies the input into either the COPD or Healthy category.

3.3.2 Training Process

a. Pretraining (Unsupervised Learning):

- Each RBM is trained individually in an unsupervised manner using contrastive divergence to learn the distribution of the input data.
- The goal is to capture meaningful representations of the data without any labeled outputs .

b. Fine-tuning (Supervised Learning):

- After pretraining, the entire DBN is fine-tuned with labeled data (COPD/Healthy).
- The backpropagation algorithm is used to minimize classification errors, adjusting weights in the network.

3.3.3 Feature Extraction for Audio Data

- Audio Preprocessing:

Cough recordings are processed to extract Mel-frequency cepstral coefficients (MFCCs), which capture the frequency and amplitude patterns of the cough.

- Feature Selection:

Features like pitch, duration, and intensity of the cough are extracted and used alongside medical data as inputs for the DBN.

3.3.4 DBN for COPD Classification

- The trained DBN processes the features through multiple layers of RBMs, refining the input data representation at each layer.
- The final output is a probability for the COPD or Healthy class, where the class with the highest probability is selected as the prediction.

3.3.5 Model Evaluation Metrics

- Accuracy: Measures the proportion of correctly classified instances.
- Precision and Recall: Evaluates how well the model identifies COPD cases (precision) and how many actual COPD cases are detected (recall).
- F1-Score: The harmonic mean of precision and recall, balancing both metrics.
- ROC-AUC: The area under the ROC curve, assessing the model's ability to discriminate between classes.

3.4 SAMPLE CODE:

```
import os
import librosa
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import tensorflow as tf
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import classification_report, confusion_matrix

DATASET_PATH = "./Respiratory_Sound_Database"
AUDIO_PATH = os.path.join(DATASET_PATH, "audio_and_txt_files").replace("\\", "/")
LABELS_FILE = os.path.join(DATASET_PATH, "patient_diagnosis.csv")

labels_df = pd.read_csv(LABELS_FILE)
labels_df.columns = ['patient_id', 'disease']
labels_df['patient_id'] = labels_df['patient_id'].astype(str)
print(labels_df)

def extract_features(file_path, max_pad_len=862):
    try:
        y, sr = librosa.load(file_path, sr=None)
        mfcc = librosa.feature.mfcc(y=y, sr=sr, n_mfcc=40)
        pad_width = max_pad_len - mfcc.shape[1]
        if pad_width > 0:
            mfcc = np.pad(mfcc, pad_width=((0, 0), (0, pad_width)), mode='constant')
        else:
            mfcc = mfcc[:, :max_pad_len]
        return mfcc
    except Exception as e:
        print(f'Error loading {file_path}: {e}')
        return None

def augment_audio(file_path, num_augmented=3):
    augmented_samples = []
    try:
        y, sr = librosa.load(file_path, sr=None)
        for _ in range(num_augmented):
            choice = random.choice(['noise', 'stretch', 'pitch'])
            if choice == 'noise':
                noise = np.random.randn(len(y))
```

```

        y_aug = y + 0.005 * noise
    elif choice == 'stretch':
        rate = random.uniform(0.8, 1.2)
        y_aug = librosa.effects.time_stretch(y, rate=rate)
    elif choice == 'pitch':
        steps = random.randint(-2, 2)
        y_aug = librosa.effects.pitch_shift(y, sr=sr, n_steps=steps)

    mfcc = librosa.feature.mfcc(y=y_aug, sr=sr, n_mfcc=40)
    pad_width = 862 - mfcc.shape[1]
    if pad_width > 0:
        mfcc = np.pad(mfcc, pad_width=((0, 0), (0, pad_width)), mode='constant')
    else:
        mfcc = mfcc[:, :862]

    augmented_samples.append(mfcc)
except Exception as e:
    print(f"Error augmenting {file_path}: {e}")
return augmented_samples

data = []
labels = []

for filename in os.listdir(audio_path):
    if filename.endswith(".wav"):
        patient_id = filename.split('_')[0]
        match = labels_df[labels_df['patient_id'] == patient_id]
        if not match.empty:
            label = match['disease'].values[0]
            file_path = os.path.join(audio_path, filename)
            feature = extract_features(file_path)
            if feature is not None:
                data.append(feature)
                labels.append(label)
            if label in ['Bronchiectasis', 'Bronchiolitis', 'Pneumonia', 'URTI', 'LRTI']:
                aug_feats = augment_audio(file_path, num_augmented=3)
                for aug in aug_feats:
                    data.append(aug)
                    labels.append(label)
from collections import Counter

label_counts = Counter(labels)
filtered_data = []

```



```

filtered_labels = []

for feature, label in zip(data, labels):
    if label_counts[label] >= 2:
        filtered_data.append(feature)
        filtered_labels.append(label)
data = np.array(filtered_data)
if data.ndim == 3:
    data = data[..., np.newaxis] # Add channel dimension only if missing

le = LabelEncoder()
labels_encoded = le.fit_transform(filtered_labels)
num_classes = len(le.classes_)

X_train, X_test, y_train_labels, y_test_labels = train_test_split(
    data, labels_encoded, test_size=0.25, random_state=42, stratify=labels_encoded
)

y_train = tf.keras.utils.to_categorical(y_train_labels, num_classes)
y_test = tf.keras.utils.to_categorical(y_test_labels, num_classes)

from tensorflow.keras import layers, models

model = models.Sequential([
    layers.Input(shape=(40, 862, 1)),
    layers.Conv2D(32, (3, 3), activation='relu', padding='same'),
    layers.MaxPooling2D((2, 2)),
    layers.BatchNormalization(),

    layers.Conv2D(64, (3, 3), activation='relu', padding='same'),
    layers.MaxPooling2D((2, 2)),
    layers.BatchNormalization(),

    layers.Reshape((10, -1)),
    layers.Bidirectional(layers.GRU(64)),
    layers.Dense(64, activation='relu'),
    layers.Dropout(0.3),
    layers.Dense(num_classes, activation='softmax')
])

model.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy'])
model.summary()
callbacks = [

```

```

        tf.keras.callbacks.ModelCheckpoint("best_model.keras", save_best_only=True,
monitor='val_accuracy', mode='max'),
        tf.keras.callbacks.EarlyStopping(patience=10, restore_best_weights=True)
]

```

```

from sklearn.utils.class_weight import compute_class_weight
class_weights = compute_class_weight(
    class_weight='balanced',
    classes=np.unique(y_train_labels),
    y=y_train_labels
)
class_weights_dict = dict(enumerate(class_weights))

```

```

history = model.fit(
    X_train, y_train,
    validation_split=0.2,
    epochs=50,
    batch_size=32,
    callbacks=callbacks,
    class_weight=class_weights_dict
)

```

```

y_pred = model.predict(X_test)
y_pred_labels = np.argmax(y_pred, axis=1)
y_true = np.argmax(y_test, axis=1)

```

```

print("\nClassification Report:\n", classification_report(y_true, y_pred_labels,
target_names=le.classes_, zero_division=0))

```

```

import seaborn as sns
conf_matrix = confusion_matrix(y_true, y_pred_labels)
plt.figure(figsize=(8, 6))
sns.heatmap(conf_matrix, annot=True, fmt="d", cmap="Blues", xticklabels=le.classes_,
yticklabels=le.classes_)
plt.ylabel('Actual')
plt.xlabel('Predicted')
plt.title('Confusion Matrix')
plt.show()

```

```

import matplotlib.pyplot as plt
plt.figure(figsize=(12, 4))
plt.subplot(1, 2, 1)
plt.plot(history.history['accuracy'], label='Train Acc')

```

```
plt.plot(history.history['val_accuracy'], label='Val Acc')  
plt.title('Accuracy')  
plt.legend()
```

```
plt.subplot(1, 2, 2)  
plt.plot(history.history['loss'], label='Train Loss')  
plt.plot(history.history['val_loss'], label='Val Loss')  
plt.title('Loss')  
plt.legend()  
plt.tight_layout()  
plt.show()
```

RESULTS AND DISCUSSIONS

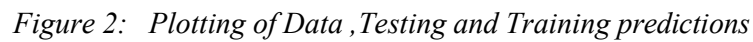


Figure 3: Training Process

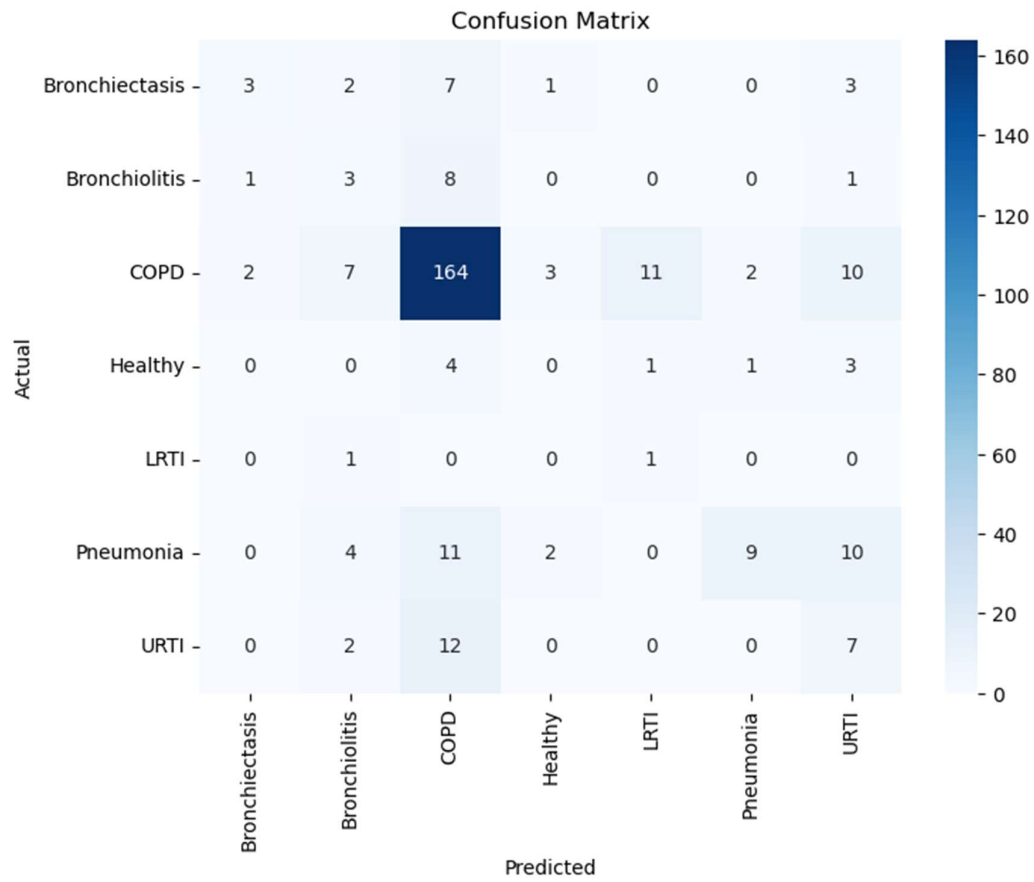


Figure 4 : Confusion Matrix

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 PROJECT CONCLUSION

The Chronic Lung Disease Predictive Analytics system using Machine Learning and Deep Belief Networks (DBN) offers a promising solution for predicting Chronic Obstructive Pulmonary Disease (COPD) and similar lung conditions. By leveraging audio data and patient-related features, the system can detect early signs of acute exacerbations and provide real-time predictions to help healthcare professionals intervene proactively.

The integration of a user-friendly chatbot and a well-structured database system allows patients to understand their condition better, track disease progression, and receive personalized recommendations. The use of supervised machine learning algorithms, such as DBN, facilitates the classification of COPD risk levels, making it an effective tool for both clinical and at-home monitoring. The backend, implemented with Python and Django, combined with a robust front-end interface, ensures accessibility for both patients and healthcare providers.

With continuous data collection and feedback, the system can evolve and improve, offering more accurate predictions over time. Moreover, the system adheres to key healthcare standards, ensuring data privacy and security, and can be integrated with wearable devices for real-time monitoring. In conclusion, this system holds the potential to revolutionize the way chronic lung diseases like COPD are managed, offering early intervention, personalized treatment plans, and a deeper understanding of disease progression.

5.2 FUTURE SCOPE

a. Integration with Wearable Devices:

Future versions of the system can integrate with wearable health-monitoring devices (such as smartwatches or IoT-based sensors) that track real-time data, including oxygen levels, heart rate, and physical activity. This continuous stream of data can provide more accurate and up-to-date predictions for COPD exacerbations.

b. Deep Learning Model Enhancement:

While the system currently uses a Multilayer Perceptron (MLP) model, there is scope for improving prediction accuracy by adopting more complex Deep Learning models such as Convolutional Neural Networks (CNNs) or Long Short-Term Memory Networks (LSTMs). These models can process sequential data more efficiently, such as cough patterns and audio features, which can lead to better identification of COPD signs.

c. Real-Time Feedback and Alerts:

The system can be upgraded to provide real-time alerts and recommendations to users or healthcare providers based on the prediction results. For example, if a sudden exacerbation is predicted, the system can immediately alert the patient to take preventative actions or seek medical attention.

d. Data Aggregation from Multiple Sources:

Future improvements can include the ability to aggregate data from various sources, such as Electronic Health Records (EHR), genetic data, and environmental factors (e.g., air pollution levels). This holistic data aggregation would allow the system to provide more comprehensive and personalized health assessments.

e. Multi-Language Support:

The chatbot and user interface could be expanded to support multiple languages, making the system accessible to a wider audience, particularly in regions where COPD prevalence is high but healthcare resources are limited.

f. Prediction for Other Respiratory Conditions:

The model can be adapted and extended to predict other respiratory diseases such as asthma or pneumonia. With proper training and additional features, the system could become a versatile tool for managing various chronic respiratory conditions.

g. Integration with Telemedicine Platforms:

Integrating the system with existing telemedicine platforms can enable remote consultations, where healthcare professionals can analyze data collected from patients' wearables and cough recordings. This can make healthcare more accessible, especially in remote areas.

CHAPTER 6

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7.	Department	Computer Science and Engineering			
8.	Details of the payment				
9.	No. of times submitted	First / Second / Third (First time – Free; Second time – Rs 200/-; Third – Rs 500/-; There after multiple of third)			
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